MATH 347: Fundamental Mathematics, Summer 2014

A.J. Hildebrand

Welcome

Welcome to Math 347! This course is an introduction to rigorous mathematical reasoning and proofs, and a preparation for upper level proof-based math courses such as Math 417 (Abstract Algebra) or Math 447 (Real Analysis). You will learn the fundamental concepts in mathematics such as sets, functions, and relations, how to construct a rigorous mathematical argument (a "proof"), and how to write up the argument in a clear and logically correct manner. Along the way, you will encounter many interesting, "cool" problems from a variety of areas of mathematics.

This course is both challenging and rewarding. It is unlike any other math course in that its focus is not on any particular mathematical subject, but rather on general mathematical concepts and abstract ideas, logical reasoning, and proofs. The emphasis is not on memorizing facts and formulas, but on learning precise mathematical thinking and logical reasoning. This course requires a serious investment of time and effort. For many students it is the most demanding math course they have taken so far. In fact, it had such a fierce reputation as a "make or break" course that there used to be T-shirts saying "I survived Math 347. And I can prove it."

This course is not for everybody, and you should not take it for the wrong reasons, e.g., because it happens to fit into your schedule, or because you need a couple more credit hours to satisfy graduation requirements. If you just want three hours of technical course work, there are many easier choices in the summer.

For those who are ready and willing to take on the challenge, the course offers many rewards. The analytical and logical skills you acquire in this course not only prepare you for advanced math courses, but are also useful in the real world. Knowing how induction *really* works, or getting comfortable with fearsome looking epsilon-delta definitions that you may have seen in calculus, but never *really* understood, can be deeply gratifying. You'll come across many interesting mathematical gems, logical puzzles, and challenging problems that one normally wouldn't see in other courses.

I will do my best to help you along the way and make this course an interesting, enjoyable, and worthwhile experience. Below is some basic information about the course; more information will be posted here as we move along. If you have any questions, concerns, or suggestions, feel free to contact me (email ajh@illinois.edu). I look forward to working with you over the coming eight weeks, and wish you an enjoyable, productive, and fulfilling, summer semester!

Course Basics

- **Course Webpage:** https://faculty.math.illinois.edu/~hildebr/347/.
- **Time/Room:** MTWRF 10:00 am - 10:50 am, W 9:00 am - 9:50 am, 141 Altgeld Hall, June 16 - August 8, 2014.
- **Instructor:** A.J. Hildebrand, office 241 Illini Hall, phone 244-7721, email ajh@illinois.edu. I will be available daily after class (in the classroom or one of the neighboring rooms), and I can set aside additional time slots later in the day if there is interest.
- **Text and syllabus:** The official text for Math 347 is "Mathematical Thinking" by D'Angelo and West (2nd edition). We will cover Chapters 1-4, 13-14, and selected topics from the other chapters depending on audience preferences. I will supplement this text with handouts summarizing the key concepts, definitions, and theorems on a given topic, and with worksheets providing additional, carefully selected practice problems.
- **Grading summary:** The course grade is based on homework assignments (30%, two assignments per week, lowest homework score dropped), in-class midterm exams (40%, 3 midterms, lowest score dropped), and a final exam (30%). For complete grading policies see the Course Information Sheet linked below.

Exams, Grades, Course Policies

- **Drop Deadline: Friday, July 11.** If you are considering dropping this course, you have to do so by this date. (If you are unsure, I'd be happy to discuss this with you!)
- **Final Exam: Friday, August 8, 2014, 8 am - 10 am.** This is the official final exam slot for this class. Please keep this date in mind when making travel plans. Instructors do not have the authority to let students take the final at a different date. In case you cannot take the final exam because of illness or another valid excuse, contact the Dean as soon as possible to discuss your options; the Dean may approve an "incomplete" grade, which will give you the opportunity to make up the final at a later date.
    - **Final Exam Syllabus:** The Final will cover the material from the three midterms in roughly equal portions (2-3 problems each), plus the Number Theory material covered during the final two weeks of class (1-2 problems). See the individual study guides for Exam 1, Exam 2, and Exam 3 for detailed syllabi on the midterm materials; and the Postexam 3 Syllabus for the material from Chapters 6 and 7 covered after the third midterm.
- **Midterm Exam 3, Wednesday, July 30.** The exam will cover the "epsilonics" material from Chapters 13 and 14. For more details see the Exam Syllabus linked below.
    - Exam Information, Syllabus, and Study Guide.
- **Midterm Exam 2, Wednesday, July 16.** The exam will cover Induction (Chapter 3) and Functions and Cardinality (Chapters 1/4). For more details see the Exam Syllabus linked below.
    - Exam Information, Syllabus, and Study Guide.
- **Midterm Exam 1, Wednesday, July 2.** The exam will cover Chapters 1 and 2, and a small part

of Chapter 3. For more details see the Exam Syllabus linked below.
    - Exam Information, Syllabus, and Study Guide.
- **Sample Exams.** Exams from past Math 347 classes.
- **Course Information Sheet.** General course information, homework and exam schedule, and complete grading policies.
- **Link to Online Scores.** Click on this link and log in with your NetID and password to access your scores. If a score is incorrect, let me know right away; any errors in the score display must be reported within a week of the assignment in question.
  Update (8/8/2014): FINAL EXAM SCORES AND COURSE GRADES ARE IN! The highest (raw) score on the final was 176/180, the median 123/180, and the average 119/180. The exam was curved in the usual way, using cutoffs of 140/115/90 for A/B/C grades. The grade shown in the online course display (including plusses and minuses) is your course grade. Enjoy the short break before the fall semester and best of luck with your future math courses!

Class Handouts and Worksheets

Class Diary

Daily class log with handouts, homework and reading assignments. **Check this page daily.**

Emergency Information.

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Class Handouts and Worksheets

Set Theory, Proof Techniques

- Review of set-theoretic notations and terminology.
- Worksheet: Set-theoretic notations and terminology.
  Solutions to Set Theory Worksheet.
- Proof techniques.
- Worksheet: Even/odd proofs.
  Solutions to "Even/odd proofs" Worksheet.
- Set-theoretic proofs.
- Worksheet: Common Errors in Proofs
  Solutions to Worksheet on Common Errors in Proof-Writing.

Logical Statements

- Logical statements: Summary
- Worksheet: Logical Statements.
  Solutions to Logic Problems Worksheet

Induction

- Worksheet: Sum/product notation.
  Solutions to "Sum/Product notation" worksheet.
- Worksheet: Induction, I: Basic Examples.
  Solutions to Induction Worksheet, I.
- Worksheet: Induction, II: Strong Induction. Recurrences and Representation
  Solutions to Induction II Worksheet.
- Worksheet: Induction, III: Miscellaneous examples.
  Solutions to Induction Worksheet III
- Worksheet: Induction, IV: Fallacies and Pitfalls.

Functions and Cardinality

- Functions: Summary of Definitions
- Worksheet: Proofs involving Functions
- Cardinality, Countable and Uncountable Sets

Epsilonics

- Sequences and Limits.
- Worksheet: Epsilonics I.
  Solutions to Epsilonics I Worksheet
- Worksheet: Epsilonics II: Infinite Series
  Solutions to Epsilonics II Worksheet
- Infinite Series: Definitions and Theorems
- Completeness Axiom and Related Properties.

Number Theory, Relations

- Number Theory: Definitions and Theorems
- Relations, Equivalence Relations, and Partitions.

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Set-theoretic Notations and Terminology

- **Sets:** A set is an **unordered** collection of objects, called the **elements** of the set. The standard notation for a set is the brace notation $\{\dots\}$, with the elements of the set listed inside the pair of braces.
- **Set builder notation:** Notation of the form $\{\dots : \dots\}$, where the colon (:) has the meaning of "such that", the dots to the left of the colon indicate the generic form of the element in the set and the dots to the right of the colon denote any constraints on the element. E.g., $\{x \in \mathbb{R} : x^2 < 4\}$ stands for *"the set of all $x \in \mathbb{R}$ such that $x^2 < 4$"*; this is the same as the interval $(-2, 2)$. Other examples of this notation are $\{2k : k \in \mathbb{Z}\}$ (set of all even integers), $\{x : x \in A \text{ and } x \in B\}$ (intersection of A and B, $A \cap B$).

  Note: Instead of a colon (:), one can also use a vertical bar (|) as a separator; e.g., $\{x \in \mathbb{R} \mid x^2 < 4\}$. Both notations are equally acceptable.
- **Equality of sets:** Two sets are equal if they contain the same elements. E.g., the sets $\{3, 4, 7\}$, $\{4, 3, 7\}$, $\{3, 3, 4, 7\}$ are all equal since they contain the same three elements, namely 3, 4, 7.
- **Some special sets:**
  - **Empty set:** $\emptyset$ (or $\{\}$)
  - **"Double bar" sets:**
    - **Natural numbers:** $\mathbb{N} = \{1, 2, \dots\}$ (note that 0 is **not** an element of $\mathbb{N}$)[1]
    - **Integers:** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
    - **Rational numbers:** $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$
    - **Real numbers:** $\mathbb{R}$
  - **Intervals:**
    - **Open interval:** $(a, b) = \{x \in \mathbb{R} : a < x < b\}$
    - **Closed interval:** $[a, b] = \{x \in \mathbb{R} : a \le x \le b\}$
    - **Half-open interval:** $[a, b) = \{x \in \mathbb{R} : a \le x < b\}$
  - **Universe:** U ("universe", a "superset" of which all sets under consideration are assumed to be a subset). (The meaning of the underlying "universe" is usually clear from the context.)
- **Some set-theoretic operations and relations:**
  - $a \in A$: "a is an element of A". Negation: $a \notin A$
  - $A \subseteq B$: "A is a subset of B". By definition, this means that every element in A is also an element in B. The negation, $A \not\subseteq B$, means that there is an element $x \in A$ with $x \notin B$. Note that the empty set, $\emptyset$, is a subset of any set, and that each set is a subset of itself.
  - $A = B$ (set equality). By definition, this means that every element in A is also an element of B, and that every element of B is also an element of A. Equivalent to "$A \subseteq B$ and $B \subseteq A$".
  - $A \cup B = \{x : x \in A \text{ or } x \in B\}$ (union)
  - $A \cap B = \{x : x \in A \text{ and } x \in B\}$ (intersection)
  - $A - B = \{x : x \in A \text{ and } x \notin B\}$ (set-theoretic difference)[2]
  - $A^c = \{x \in U : x \notin A\}$ (complement)[3]
  - $P(A) = \{B : B \subseteq A\}$ (power set of A)
  - $A \times B = \{(a, b) : a \in A, b \in B\}$ (Cartesian product)

[1] We follow here the standard convention in the English mathematical literature where 0 is not considered a natural number. In some other fields (e.g., computer science) and cultures (e.g., French) it is common to include 0 in the definition of $\mathbb{N}$.

[2] An alternate notation for the set-theoretic difference $A - B$ is $A \setminus B$ (with a backslash).

[3] Another common notation for the complement is $\overline{A}$ instead of A^c.

- **Venn diagrams:** Venn diagrams can be very helpful to get a "feel" for set operations, relations, and identities such as De Morgan's Law. *However, a Venn diagram does not constitute a proof!* For a refresher on Venn diagrams see the Wikipedia page "Venn diagrams". (Google the term.)

- **Notes on set-theoretic notation:** Normal arithmetic operations like addition, subtraction, multiplication don't make sense in the context of sets. For example, $A + B$ does not make sense, so do not use this notation (you probably meant $A \cup B$). An exception is the minus sign $(-)$, which is used for the set-theoretic difference $A - B$. However, in set-theoretic context the minus sign has a completely different meaning than the ordinary arithmetic minus sign, and properties for the arithmetic minus sign do not, in general, hold for the set-theoretic minus.

- **Further resources:**

  - D'Angelo/West text, pp. 6–10.
  - Wikipedia pages on "Set (mathematics)" and "Venn diagrams". (Google the terms.) Great as a quick refresher on basic set-theoretic concepts.
  - "Discrete Mathematics" by Rosen (on course reserve in the Math Library). This text (which is the text for Math 213 and CS 173) is a great source for additional examples and practice problems.

Number Theory: Summary of Definitions and Theorems

Divisibility:

Given arbitrary integers a, b, with $a \neq 0$, we say that **"a divides b"** if there exists $k \in \mathbb{Z}$ such that $b = ka$. If a divides b, we say b is **divisible** by a, and we call a a **divisor** (or **factor**) of b, and b a **multiple** of a.
We write $a \mid b$ for "a divides b", and $a \nmid b$ for its negation, "a does *not* divide b".
Note: The "divisor" a in this definition can be negative, but must be nonzero; divisibility by 0 is not defined.

Congruences:

Given arbitrary (positive, zero, or negative) integers a, b, and a **modulus** m, we say (**"a is congruent to b modulo m"**), and write $\boxed{a \equiv b \mod m}$ if $a = b + km$ for some integer k, or equivalently, if $m \mid a - b$
Note: The modulus m is an essential part of the definition. Make sure to always specify the modulus; saying "a is congruent to b", or writing "$a \equiv b$", without specifying a modulus makes no sense.

Properties of congruences:

What makes congruences so useful is that, to a large extent, they can be manipulated like ordinary equations. Congruences *to the same modulus* can be added, multiplied, and taken to a fixed positive integral power; i.e., for any $a, b, c, d \in \mathbb{Z}$ and $m \in \mathbb{N}$ we have:

- **Sums:** If $a \equiv b \mod m$ and $c \equiv d \mod m$, then $a + c \equiv b + d \mod m$.
- **Products:** If $a \equiv b \mod m$ and $c \equiv d \mod m$, then $ac \equiv bd \mod m$.
- **Powers:** If $a \equiv b \mod m$ and $k \in \mathbb{N}$, then $a^k \equiv b^k \mod m$.
- **Polynomial values:** If $a \equiv b \mod m$ and $P(x)$ is a polynomial with integer coefficients, then $P(a) \equiv P(b) \mod m$.

Primes and composite numbers:

An integer n with $n \geq 2$ is called a **prime** if its only *positive* divisors are 1 and n, and **composite** otherwise. Equivalently, n is composite if it be factored in the form $n = ab$ with integers a, b satisfying $1 < a, b < n$
Note: The natural number 1 is considered neither prime nor composite. Negative numbers and 0 are not classified as prime or composite either.

Fundamental Theorem of Arithmetic:

$$\boxed{\text{Every integer } n \geq 2 \text{ has a unique factorization into primes.}}$$

That is, every integer $n \geq 2$ can be represented in the form $n = p_1 p_2 \dots p_r$, where the p_i are primes (not necessarily distinct). Moreover, this representation is unique except for the ordering of the primes p_i.

Euclid's Theorem:

$$\boxed{\text{There are infinitely many prime numbers.}}$$

This is perhaps the most famous theorem in Number Theory. Euclid's proof, given below, is a classic, one of the most memorable proofs in all of mathematics, and a wonderful illustration of the method of contradiction. This is a proof that is worth remembering for the rest of your life!
Proof of Euclid's Theorem: We use contradiction. Assume there are only finitely many prime numbers, say $p_1, p_2, \dots, p_n$. Let $N = p_1 p_2 \dots p_n + 1$. Then N is an integer ≥ 2, so by the Fundamental Theorem of Arithmetic it can be written as a product of one or more prime numbers. Since, by our assumption, $p_1, \dots, p_n$ are all the primes, at least one of these, say p_i, must appear in the factorization of N, and so $p_i \mid N$. On the other hand, $p_i \mid p_1 p_2 \dots p_n = N - 1$. Since $d \mid a$ and $d \mid b$ implies $d \mid a - b$, it follows that $p_i \mid N - (N - 1) = 1$, which is impossible, since $p_i \geq 2$. This contradiction proves the claim.

Further Resources:

In the text this material can be found in Chapters 6 and 7. A fantastic resource on everything about primes is the Prime Pages website, `http://primes.utm.edu`.

Relations, Equivalence Relations, and Partitions

[See also the beginning of Chapter 7, pp. 140–141, of the text.]

Relations

- A **relation from a set S to a set T** is a subset of $S \times T$.
- A **relation on a set S** is a relation from S to S, i.e., a subset of $S \times S$.
- **Notation:** Given a relation R on S (i.e., a subset $R \subseteq S \times S$), we write "$x \sim y$" if $(x, y) \in R$. A relation can be described by specifying the meaning of "$x \sim y$".
- **Example: Congruence relation modulo m.** Given a positive integer m, the congruence relation modulo m is the relation on the set $S = \mathbb{Z}$ defined by $\boxed{x \sim y \Longleftrightarrow x \equiv y \mod m}$, or equivalently, by the subset $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \equiv y \mod m\}$ of $\mathbb{Z} \times \mathbb{Z}$.

Equivalence relations and equivalence classes

- **Equivalence relation on a set S:** A relation on S that satisfies the following properties:
  - **Reflexive:** For all $x \in S$, $x \sim x$.
  - **Symmetric:** For all $x, y \in S$, $x \sim y$ implies $y \sim x$.
  - **Transitive:** For all $x, y, z \in S$, $x \sim y$ and $y \sim z$ implies $x \sim z$.
- **Equivalence classes:** If $\sim$ is an equivalence relation on S and $x \in S$, the set
$$E_x = \{y \in S : x \sim y\}$$
is called the **equivalence class of x**. Other notations for the equivalence class E_x of x are the bracket and bar notations, $[x]$ and, $\overline{x}$.
- **Example: Congruence relation modulo 3.** The congruence relation modulo 3 is an equivalence relation with the following equivalence classes:
$$E_0 = \overline{0} = \{0, 3, 6, 9, \dots, -3, -6, -9, \dots\},$$
$$E_1 = \overline{1} = \{1, 4, 7, 10, \dots, -2, -5, -8, \dots\},$$
$$E_2 = \overline{2} = \{2, 5, 8, 11, \dots, -1, -4, -7, \dots\}.$$
Every integer belongs to exactly one of these three classes, and any other equivalence class $\overline{x}$ is equal to one of the above classes. For example, the equivalence class of 3, i.e., $\overline{3}$, is the same as $\overline{0}$.

Partitions of a set

Let S be a set. A collection of (finitely or infinitely many) nonempty subsets $A_1, A_2, \dots$ of S is called a **partition of S** if it has the following properties:

- **No overlaps:** The sets A_i are pairwise disjoint; i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$.
- **Union is all of S:** The union of all the sets A_i in the partition is S, i.e., $A_1 \cup A_2 \cup A_3 \cup \cdots = S$.

Equivalence relations and partitions: The Fundamental Theorem

The following fundamental result connects the seemingly unrelated concepts of an equivalence relation on a set S (which, formally, is a subset of $S \times S$ with certain properties) and a partition on S (which is a collection of subsets of S with certain properties).

- **From equivalence relations to partitions:** Given an equivalence relation on a set S, **the set of distinct equivalence classes forms a partition of S**. That is, we have: (i) For any $x, y \in S$, the associated equivalence classes E_x and E_y are either equal (i.e., $E_x = E_y$) or disjoint (i.e., $E_x \cap E_y = \emptyset$). (ii) For any $x \in S$ there exists an equivalence class that contains x (namely, the class E_x).
- **From partitions to equivalence relations:** Given a partition of S into sets $A_1, A_2, \dots, A_n$, the relation defined by $\boxed{\text{"}x \sim y\text{"} \Longleftrightarrow \text{"}x \text{ and } y \text{ belong to the same set } A_i \text{ from the partition"}}$ is an equivalence relation on S.